ST2MML

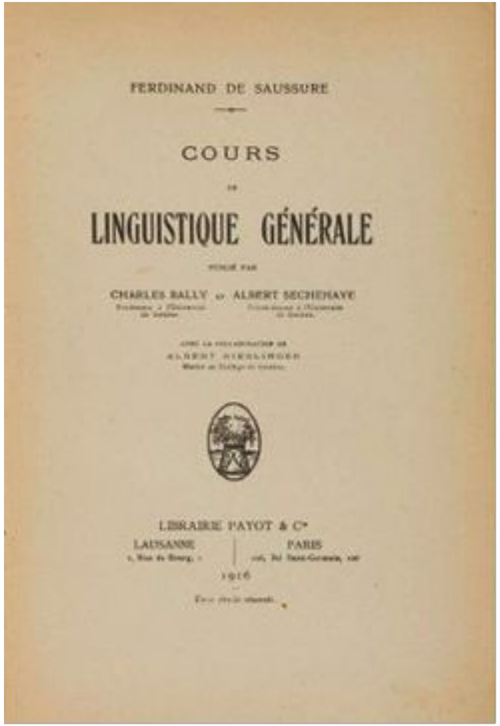
NLP

Today

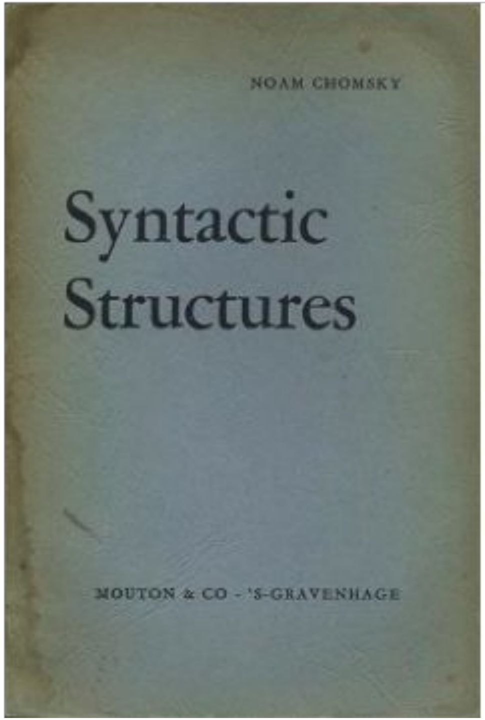
- Classic NLP
    - some libraries: spacy, NLTK
- Similarity and Embeddings
- Demo : sentence similarity with wikipedia
- OpenAI API
-

Classic
NLP

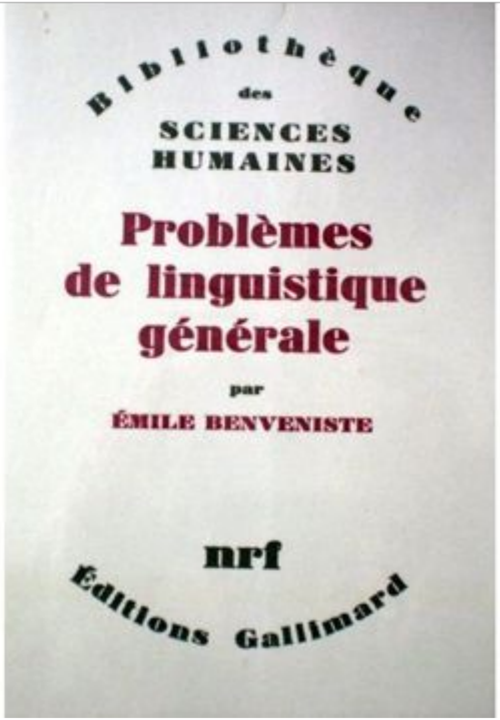
From linguistique to NLP



1916 Ferdinand de Saussure

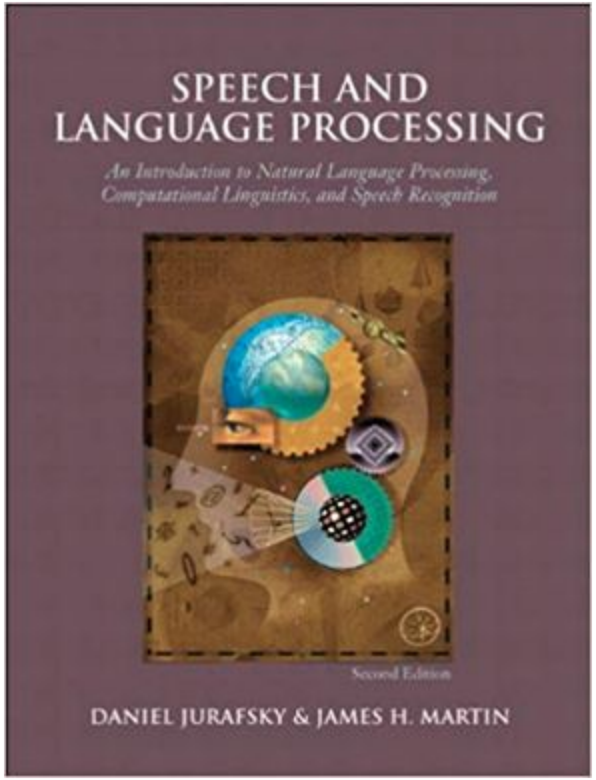


1966 Chomsky



1957 Benveniste

Book of reference



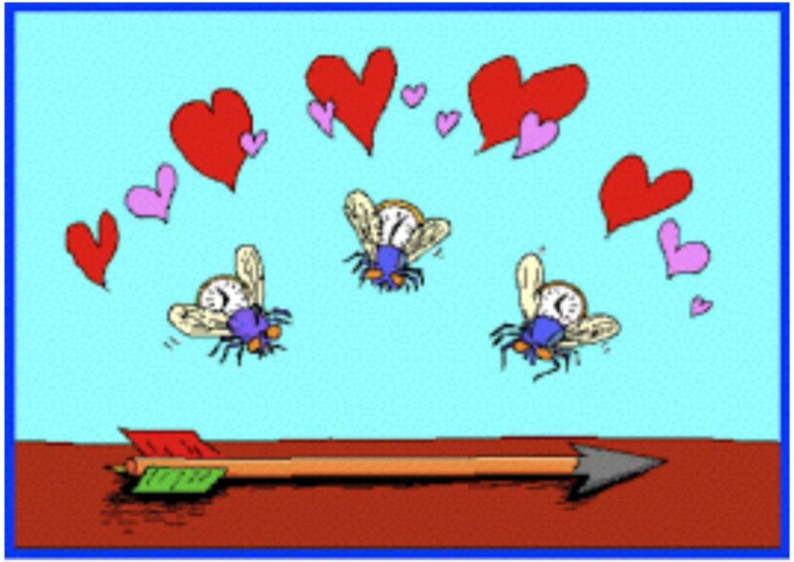
[Speech and Language Processing (3rd ed. draft) Dan Jurafsky and James H. Martin]

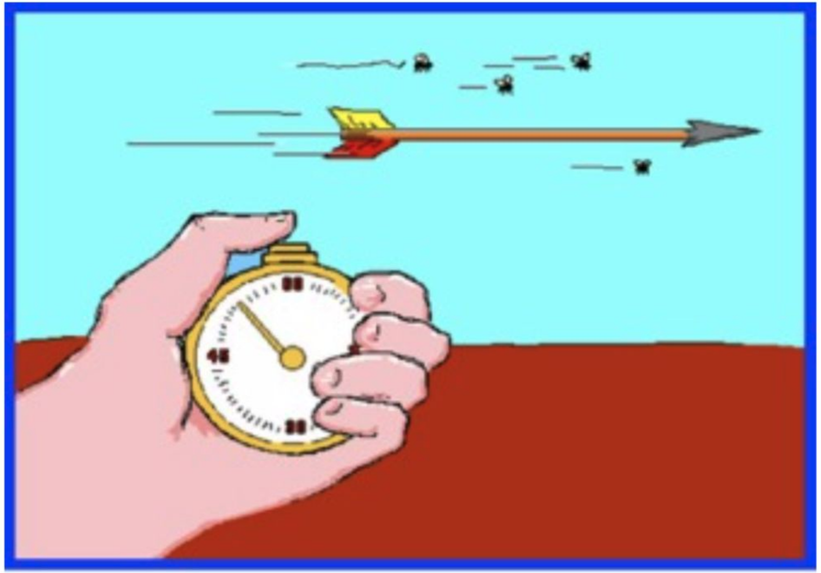
Ambiguities

- "Will you marry me?" is a marriage proposal.
- "Will, You, Mary, Me" is a card game proposal.
- "Will, you marry me" is a time traveller spoiling the future.
- "Will you, Mary me" is a cavewoman named Mary, trying to make Will, who has amnesia, remember who he is.

- I saw her duck
- I saw her duck. She tried to avoid a flying ball.

- Let's eat, grandpa.
- Let's eat grandpa.





Time flies like an arrow

Applications - classic NLP

- Text mining
    - NER: Named Entity Recognition: LOC, PER,
    - POS: Part of speech Tagging : nouns, adjectives, verbs, …
- speech to text, text to speech
- Translation
- Classification: sentiment analysis, spam, ...
- Text generation, summarization, question answering
- Topic identification, topic modeling
- word sense disambiguation : bank,

Written text is extremely complex

- Non standard language
- Variable length
- Discrete: rat <> sat, while image and sound are continuous
- Wide variety of complexity across languages
    - Arabic: ...
    - German: Donaudampfschiffahrtsgesellschaftskapitän (5 “words”)
    - Chinese: 50,000 different characters (2-3k to read a newspaper)
    - Japanese: 3 writing systems
    - Thai: Ambiguous word boundaries and sentence concepts
    - Slavic: Different word forms depending on gender, case, tense
- Encoding: unicode vs Ascii
- **unstructured data**

Wayyyyyy back in 2021! 4 years ago!

Easy

- Chunking
- Part-of-Speech Tagging
- Named Entity Recognition
- Spam Detection
- Thesaurus

Medium

- Syntactic Parsing
- Word Sense Disambiguation
- Sentiment Analysis
- Topic Modeling
- Information Retrieval

Hard

- Machine Translation
- Text Generation
- Automatic Summarization
- Question Answering
- Conversational Interfaces

Demo classic NLP

https://cloud.google.com/natural-language

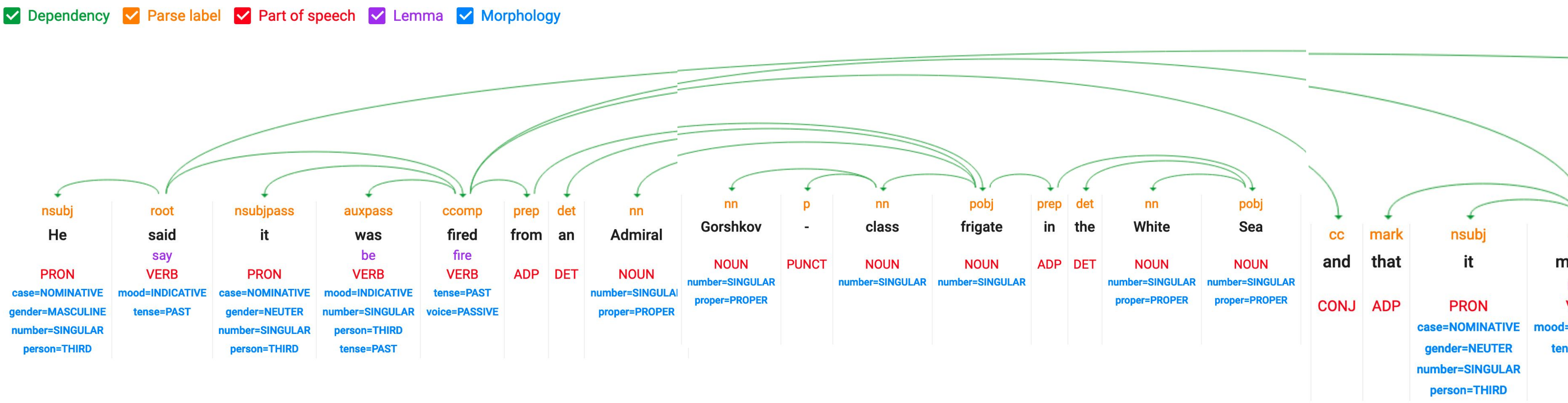
Valery Gerasimov, chief of the army's general staff, said the Tsirkon missile covered a distance of more than 280 miles in 4.5 minutes to destroy a target in the Barents Sea off Russia's Arctic coast. He said it was fired from an Admiral Gorshkov-class frigate in the White Sea and that it marked the first time the missile had hit a target at sea.

From Russia tests hypersonic missile on Putin's birthday | World

SpaceX began a new era of spaceflight: For all those searching for a new planet to call home, the year brought at least a bit of good news. SpaceX, the company founded by billionaire Elon Musk to fulfill his dream of colonizing Mars, launched NASA astronauts into orbit for the first time since the U.S. government retired the space shuttle program in 2011. SpaceX regularly transports cargo to the International Space Station, and in 2020 became the first private enterprise ever to launch astronauts there.

From 2020 Events - Pop Culture, U.S. Politics & World | HISTORY

Part of Speech tagging



Stopwords

- many words have no meaning: stopwords

```
stopwords = ['the', 'of', 'and', 'is','to','in','a','from','by','that',
'with', 'this', 'as', 'an', 'are','its', 'at', 'for', …]
```

What is the unit ?

We can work with

- words, syllables, tokens, letters & punctuation
- **Bigrams**, **n-grams**: New York, cul-de-sac, pain au chocolat
- sentences, paragraphes, tweets, articles, books, comments
- Corpus: a whole set of text

And

- The vocabulary is large / infinite and fast changing
- Typos, multiple spellings

Vocabularies

- fixed and large size : balance between performance and exhaustivity / meaning
- Out of Vocabulary
  - Named entities and proper nouns
  - Domain-specific technical terms
  - Neologisms and emerging words
  - Misspellings and typos
  - Morphological variants

Tokenization: defining the unit of text

- What's the most efficient unit of text ?
- Using words is problematic:
  - very large vocabulary, multiple forms: plural, declension (home, house), conjugation, ...,etc
- Using letters is too short

Let's split words into multiple tokens: **subwords** tokenization

- unhappiness -> un + happ + iness
- running -> run + ing
- biiig -> b + iii + g

Benefit of subwords tokenization

- no words are OOV: out of vocabulary
    - gpt 3.5 did not know about COVID
- typo resilient
- smaller vocabulary size
- captures semantic and morphological meaning better

LLM pricing - based on input - output Tokens

OpenAI o1

Frontier reasoning model that supports tools, Structured Outputs, and vision | 200k context length

Price

Input:
$15.00 / 1M tokens

Cached input:
$7.50 / 1M tokens

Output:
$60.00 / 1M tokens

OpenAI o3-mini

Small cost-efficient reasoning model that's optimized for coding, math, and science, and supports tools and Structured Outputs | 200k context length

Price

Input:
$1.10 / 1M tokens

Cached input:
$0.55 / 1M tokens

Output:
$4.40 / 1M tokens

Pricing | OpenAI Feb 4th 2025

Embeddings

A distance between words

How to calculate a distance between 2 texts ?

so that we can say that

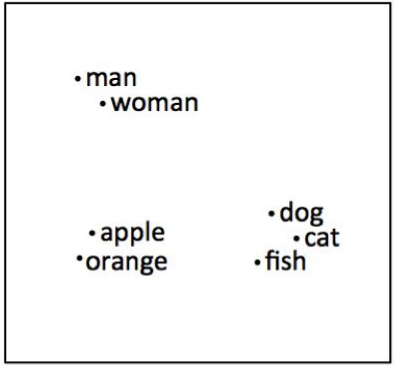
- **banana** is closer to **apple** than it is to **plane**;
- `the dog barks` is closer to `the cat meows` than it is to `the tractor sputters`, …

before 2013: counting words and their relative frequencies
(messy, crude, called tf-idf, worked well for spam detection)

Word2Vec

2013: Thomas Mikolov @Google Efficient Estimation of Word Representations in Vector Space https://arxiv.org/abs/1301.3781

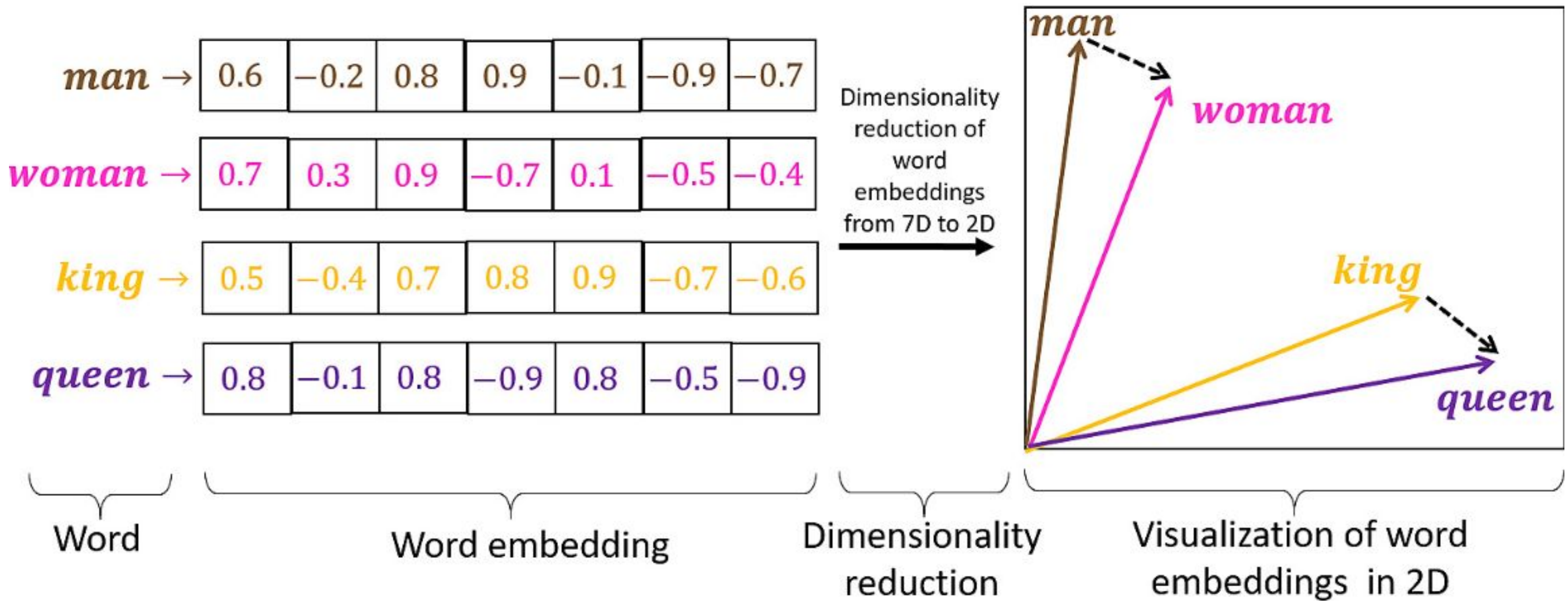
- Queen - Woman = King - Man
- France + capital = Paris



no context disambiguation: bank (river) = bank (finance)

2013! Word2vec

One word has one vector of large dimension

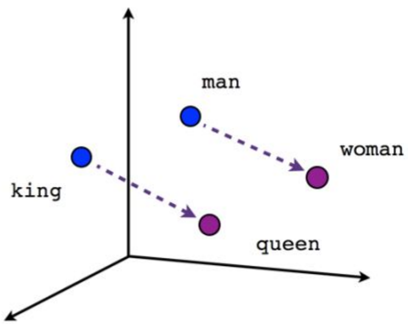


https://airbyte.com/data-engineering-resources/sentence-word-embeddings

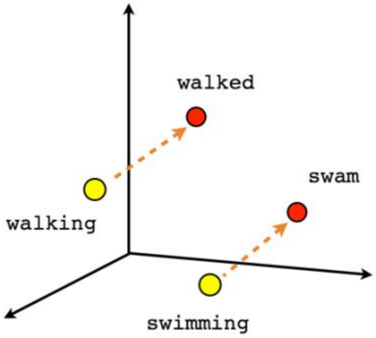
distance with Cosine Similarity

Once you have embeddings of 2 texts (sentences, words, tokens ,...)
The distance between the 2 text is given by the cosine similarity of their embeddings

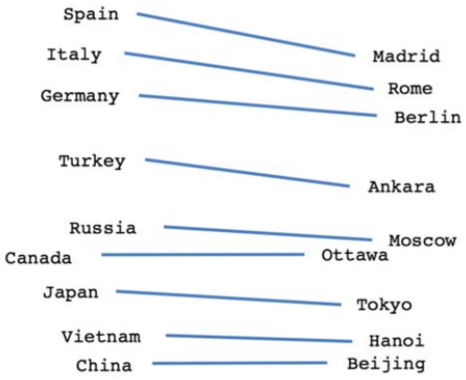
$$sim(u, v) = \frac{u^T v}{\|u\|_2 \|v\|_2}$$



Male-Female



Verb tense



Country-Capital

Current embeddings methods

Embedding using an LLM

Hugging Face Transformers: For models like BERT, RoBERTa, SBERT.

OpenAI API: For models like text-embedding-ada-002.

Example Workflow:

1. Load a pre-trained SBERT model.
2. Pass your sentences through the model.
3. Get a 768-dimensional vector (embedding) for each sentence.
4. Use these embeddings for your task.

Demo

Find immigration related text in wikipedia pages of main European capitals

- use chatGPT project
- use wikipedia API
- get the pages for Paris, Rome, Berlin, …
- split the text into sentences
- calculate the embeddings for each sentence
- given a query sentence and its embeddings
    - find the N sentences that are most similar to the query sentence
- save the corpus

https://colab.research.google.com/drive/16wQ3o8XQNEhADR2v2ibwfvjaKXKvkyKW#scrollTo=cGm37GYrSNd9

OpenAI API

The secret key

- create an account
- provision some money (~$2, $3)
- create a secret key
  - save that secret key in a safe place on your computer
  - never put the key in the code
  - in colab, add the key on the left side
- for OpenAI the name of the key is
  - `OPENAI_API_KEY`